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| Programme      | B. Sc. Computer Science                                                                                                                                                                                                                                                                                                          |                  |                   |                    |             |
| Course Title   | Data Analysis and Visualisation Using Spreadsheets                                                                                                                                                                                                                                                                               |                  |                   |                    |             |
| Type of Course | Vocational Minor                                                                                                                                                                                                                                                                                                                 |                  |                   |                    |             |
| Semester       | III                                                                                                                                                                                                                                                                                                                              |                  |                   |                    |             |
| Academic Level | 200-299                                                                                                                                                                                                                                                                                                                          |                  |                   |                    |             |
| Course Details | Credit                                                                                                                                                                                                                                                                                                                           | Lecture per week | Tutorial per week | Practical per week | Total Hours |
|                | 4                                                                                                                                                                                                                                                                                                                                | 3                | -                 | 2                  | 75          |
| Pre-requisites | <ul style="list-style-type: none"> <li>• Basic understanding of computers</li> <li>• Familiarity with basic mathematical operations</li> </ul>                                                                                                                                                                                   |                  |                   |                    |             |
| Course Summary | This course provides a comprehensive introduction to Spreadsheets, focusing on understanding formulas, functions, data organization, analysis techniques, and data visualization. Participants will gain skills in spreadsheet management, data cleansing, analysis, and visualization using Excel's various tools and features. |                  |                   |                    |             |

#### Course Outcomes (CO):

| CO  | CO Statement                                                                                                                                                                                                                                                                                                                      | Cognitive Level* | Knowledge Category# | Evaluation Tools used                         |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------------|-----------------------------------------------|
| CO1 | Students will demonstrate proficiency in managing spreadsheets, including creating, formatting, and manipulating data within Excel workbooks. They will be able to effectively navigate Excel's interface and utilize toolbars.                                                                                                   | U                | P                   | Instructor-created exams / Quiz               |
| CO2 | Learners will understand the importance of data organization and cleansing in Excel. They will be able to import, export, filter, sort, validate, and remove duplicates from datasets. Students will develop skills to ensure data integrity and consistency, enhancing their ability to work with clean and organized data sets. | U                | P                   | Instructor-created exams/<br>Home Assignments |
| CO3 | Participants will acquire advanced data analysis skills like pivot tables, what-if analysis, and goal seek. They will be able to apply various Excel functions and tools to perform complex calculations, analyze trends, and make informed                                                                                       | Ap               | P                   | Instructor-created exams                      |

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|                                                                                                                                                                                                   | decisions based on data analysis.                                                                                                                                                                                                                                                                                    |    |   |                          |
| CO4                                                                                                                                                                                               | Students will gain proficiency in data visualization techniques using Excel. They will be able to create a variety of charts, design pivot charts, dashboards for effective data analysis. Additionally, learners will be able to implement form controls for interactive data manipulation in their visualizations. | Ap | P | Instructor-created exams |
| CO5                                                                                                                                                                                               | Learners will develop skills in advanced features of Excel like macros, protect data sheets and workbooks, utilize split, freeze, and hide options effectively, incorporate add-ins for extended functionalities, and manage printing options in Excel for professional presentation of data.                        | Ap | P | Instructor-created exams |
| * - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)<br># - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P)<br>Metacognitive Knowledge (M) |                                                                                                                                                                                                                                                                                                                      |    |   |                          |

### Detailed Syllabus:

| Module     | Unit                                                                 | Content                                                                                                                                                         | Hrs<br>(45) | Marks<br>(70) |
|------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------|
| <b>I</b>   | <b>Introduction to Excel &amp; Understanding Formulas, Functions</b> |                                                                                                                                                                 | <b>10</b>   | <b>15</b>     |
|            | 1                                                                    | Features of Spreadsheet                                                                                                                                         | 1           |               |
|            | 2                                                                    | Parts of Excel Window, Toolbars, Worksheet and Workbook, Insertion and Deletion of cells, columns, rows                                                         | 2           |               |
|            | 3                                                                    | Formatting in Excel (Merge, Warp, Font Formatting, Number Formatting, Borders and Shading, Colouring)                                                           | 2           |               |
|            | 4                                                                    | Range, Autofill, Autosum, Relative, Absolute and Mixed Referencing in Excel, Linking data between worksheets                                                    | 2           |               |
|            | 5                                                                    | Formulas and Functions in Excel: Use of Formula Bar, Functions: SUM, ROUND, CEIL, FLOOR, IF, AND, OR, AVERAGE, MIN, MAX ,COUNT, COUNTIF, SUMIF, VLOOKUP,HLOOKUP | 3           |               |
| <b>II</b>  | <b>Cleansing and Organising Data in Excel</b>                        |                                                                                                                                                                 | <b>10</b>   | <b>15</b>     |
|            | 6                                                                    | Importance of Data Cleansing and Organisation                                                                                                                   | 1           |               |
|            | 7                                                                    | Data Import and Export                                                                                                                                          | 2           |               |
|            | 8                                                                    | Filtering and Sorting                                                                                                                                           | 2           |               |
|            | 9                                                                    | Data Validation and removal of duplicates                                                                                                                       | 2           |               |
|            | 10                                                                   | Group, Ungroup, Subtotal                                                                                                                                        | 2           |               |
|            | 11                                                                   | Conditional Formatting – Highlight Cell Rules, Top/Bottom Rules                                                                                                 | 1           |               |
| <b>III</b> | <b>Advanced Techniques for Data Analysis</b>                         |                                                                                                                                                                 | <b>14</b>   | <b>20</b>     |
|            | 12                                                                   | Features of Pivot table                                                                                                                                         | 1           |               |

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|           | 13                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Pivot Table creation                                                                                                | 2         |           |
|           | 14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Fitting Linear regression in Excel                                                                                  | 3         |           |
|           | 15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Linear regression using Excel formulas                                                                              | 3         |           |
|           | 16                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Interpreting regression results                                                                                     | 2         |           |
| <b>IV</b> | <b>Data Visualisation Techniques</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                     | <b>14</b> | <b>20</b> |
|           | 17                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Creating Charts, Different types of charts                                                                          | 2         |           |
|           | 18                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Formatting Chart Objects, Changing the Chart Type, Showing and Hiding the Legend, Showing and Hiding the Data Table | 2         |           |
|           | 19                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Creating charts from regression                                                                                     | 2         |           |
|           | 20                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Pivot Chart                                                                                                         | 2         |           |
|           | 21                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Dashboards                                                                                                          | 2         |           |
|           | 22                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Form Controls                                                                                                       | 4         |           |
| <b>V</b>  | <b>Hands-on Spreadsheets</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                     | <b>30</b> |           |
|           | <b>Spreadsheet Basics:</b> <ol style="list-style-type: none"> <li>1. Create a new workbook in Excel.</li> <li>2. Identify and label different parts of the Excel window, such as the Ribbon, Formula Bar, Name Box, and Worksheet Tabs.</li> <li>3. Insert and delete cells, columns, and rows within a worksheet.</li> </ol>                                                                                                                                                                              |                                                                                                                     |           |           |
|           | <b>Formatting:</b> <ol style="list-style-type: none"> <li>4. Merge cells and wrap text within merged cells.</li> <li>5. Experiment with different font styles, sizes, and colors for text formatting.</li> <li>6. Apply various number formatting options (e.g., currency, percentage, date) to cells.</li> <li>7. Add borders and shading to cells or ranges.</li> </ol>                                                                                                                                  |                                                                                                                     |           |           |
|           | <b>Range Operations:</b> <ol style="list-style-type: none"> <li>8. Use Autofill to quickly populate a series of cells with data (e.g., numbers, dates, text).</li> <li>9. Utilize Autosum to calculate the sum of a range of numbers automatically.</li> <li>10. Practice relative, absolute, and mixed referencing in formulas to understand their impact on cell references.</li> <li>11. Link data between different worksheets within the same workbook.</li> </ol>                                    |                                                                                                                     |           |           |
|           | <b>Formulas and Functions:</b> <ol style="list-style-type: none"> <li>12. Experiment with different mathematical formulas (e.g., addition, subtraction, multiplication, division) using the Formula Bar.</li> <li>13. Apply common functions such as SUM, ROUND, CEIL, FLOOR, IF, AND, OR, AVERAGE, MIN, MAX, COUNT, COUNTIF, SUMIF, VLOOKUP, and HLOOKUP to solve specific problems or analyze data sets.</li> <li>14. Combine functions within formulas to perform more complex calculations.</li> </ol> |                                                                                                                     |           |           |
|           | <b>Data Import and Export:</b> <ol style="list-style-type: none"> <li>15. Import external data from sources such as CSV files, text files, or databases into Excel.</li> <li>16. Export Excel data to different formats (e.g., CSV, PDF) for sharing or further analysis.</li> </ol>                                                                                                                                                                                                                       |                                                                                                                     |           |           |
|           | <b>Filtering and Sorting:</b> <ol style="list-style-type: none"> <li>17. Filter data to display specific records based on criteria (e.g., dates,</li> </ol>                                                                                                                                                                                                                                                                                                                                                |                                                                                                                     |           |           |

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|  | categories, numerical ranges).<br>18. Sort data alphabetically, numerically, or chronologically to analyze trends or identify patterns.                                                                                                                                                                                                                                                                 |  |  |
|  | <b>Data Validation and Removal of Duplicates:</b><br>19. Implement data validation rules to restrict input values within specified criteria (e.g., date ranges, numerical limits, list selections).<br>20. Identify and remove duplicate records from a dataset while preserving unique data entries.                                                                                                   |  |  |
|  | <b>Grouping, Ungrouping, and Subtotal:</b><br>21. Group related rows or columns together to organize data hierarchically.<br>22. Perform subtotal calculations within grouped data to summarize information at different levels of detail.                                                                                                                                                              |  |  |
|  | <b>Conditional Formatting:</b><br>23. Apply conditional formatting using highlight cell rules to visually identify data outliers, trends, or exceptions.<br>24. Utilize top/bottom rules to highlight top or bottom values within a dataset for quick analysis.                                                                                                                                         |  |  |
|  | <b>Pivot Table Creation:</b><br>25. Import a dataset into Excel and create a pivot table summarizing key metrics (e.g., sales revenue, product quantities) by different dimensions (e.g., region, product category).<br>26. Experiment with different pivot table configurations (e.g., adding calculated fields, grouping data, creating hierarchical rows/columns) to gain insights into the dataset. |  |  |
|  | <b>Fitting Linear Regression in Excel:</b><br>27. Import a dataset containing variables for linear regression analysis (e.g., independent and dependent variables).<br>28. Use Excel's built-in regression analysis tool to fit a linear regression model to the data and calculate coefficients, standard errors, and goodness-of-fit measures.                                                        |  |  |
|  | <b>Creating Charts from Regression Analysis:</b><br>29. Perform linear regression analysis on a dataset containing independent and dependent variables.<br>30. Create a scatter plot chart to visualize the relationship between the variables, including the regression line and confidence intervals.                                                                                                 |  |  |
|  | <b>Pivot Chart:</b><br>31. Create a pivot chart based on a pivot table summarizing key metrics from a dataset.<br>32. Customize the pivot chart to display data trends and patterns dynamically as the underlying pivot table data is updated.                                                                                                                                                          |  |  |
|  | <b>Dashboards:</b><br>33. Design a dashboard incorporating multiple charts and pivot tables to provide a comprehensive overview of business metrics or performance indicators.<br>34. Use interactive features such as slicers and timeline controls to enable users to filter and analyze data dynamically.                                                                                            |  |  |
|  | <b>Form Controls:</b><br>35. Add form controls such as checkboxes, dropdown lists, and option buttons to interact with charts and pivot tables.                                                                                                                                                                                                                                                         |  |  |

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|  | 36. Create interactive features allowing users to customize chart views or update data dynamically based on user inputs. |  |  |
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### **References**

1. "Excel 2019 Bible" by Michael Alexander and Richard Kusleika
2. "Excel Formulas & Functions For Dummies" by Ken Bluttman and Peter Aitken
3. “Excel with Microsoft Excel: Comprehensive & Easy Guide to Learn Advanced MS Excel” by Naveen Mishra

### **Assessment Rubrics:**

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Final Exam